UNIT - I

INTRODUCTION TO TQM

Quality - Dimensions - TQM - Quality Planning - Quality Costs - Basic Concepts

1. Definition of Quality

Quality = fitness for use (Juran). The product or service should satisfy the needs of the customer at a price he is willing to pay.

- Latin word "**qualitas**" meaning "nature of a thing" or "degree of excellence".
- Crosby: Quality is **conformance to requirements** (zero defects).
- Deming: Quality is **a predictable degree of uniformity and dependability at low cost,** suited to the market.
- Feigenbaum: Quality is the total composite product and service characteristics of marketing, engineering, manufacturing and maintenance through which the product meets the expectation of the customer.
- ISO 9000:2000: Quality is the **degree to which a set of inherent characteristics fulfils requirements.**
- Taguchi: Quality is the loss imparted to society from the time the product is shipped.
- Customer's view: **quality = meeting or exceeding customer expectations.**

Quality can be viewed in 5 ways

- **Transcendent view:** quality is innate excellence, recognised only by experience (e.g. a Rolls Royce).
- **Product based view:** quality is a measurable quantity - more

features/ingredients means higher quality.

- **User based view:** "fitness for use" - depends on who the user is.
- **Manufacturing based view:** conformance to specifications, prevent defects, reduce rework.
- **Value based view:** performance at an acceptable price, quality + cost together.

Why is quality important?

- Customer satisfaction and loyalty, repeat business.
- Competitive advantage, global market share.
- Lower cost - less scrap, rework and warranty claims.
- Company reputation and image, higher profit.
- Product liability - poor quality can lead to legal action.

> Remember: Quality is not an expense, it is an investment. Quality is everybody's job.

2. Dimensions of Quality

Two sets are commonly written. Garvin's 8 dimensions for **products** and the 5 dimensions (RATER) for **services.**

A. Garvin's eight dimensions of product quality

1. **Performance:** primary operating characteristics of a product. Eg. picture clarity of a TV, speed of a car.
2. **Features:** secondary characteristics that supplement the basic function. Eg. remote control, Bluetooth in a car.
3. **Reliability:** probability that the product will not fail within a given time. Measured by MTBF (mean time between failures).
4. **Conformance:** degree to which design and operating characteristics meet established standards/specifications.
5. **Durability:** amount of use before the product deteriorates or must be replaced - the product life.

6. **Serviceability:** speed, courtesy, competence and ease of repair; after-sales service.
7. **Aesthetics:** how the product looks, feels, sounds, tastes or smells - personal judgement.
8. **Perceived quality:** reputation, brand image, advertising - indirect measures of quality.

B. Dimensions of service quality (SERVQUAL - Parasuraman)

Dimension	Meaning	Example
Tangibles	physical facilities, equipment, appearance of staff	clean bank branch, uniforms
Reliability	perform promised service dependably and accurately	order delivered on time
Responsiveness	willingness to help, prompt service	quick reply to a complaint
Assurance	knowledge, courtesy, ability to inspire trust	doctor's confidence
Empathy	caring, individual attention to customer	personalised attention

Other service dimensions

- Time (waiting time), Timeliness, Completeness, Courtesy, Consistency, Accessibility and convenience, Accuracy, Responsiveness.

Exam tip: write Garvin's 8 as the mnemonic "P F R C D S A P" (Performance, Features, Reliability, Conformance, Durability, Serviceability, Aesthetics, Perceived quality).

3. Definition of Total Quality Management

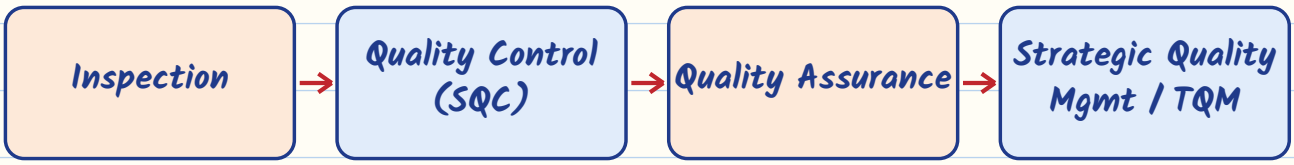
- **Total:** involves everyone in the organisation - top management to workers, suppliers and customers.
- **Quality:** meeting customer requirements, fully and every time.

- **Management:** planning, organising, leading and controlling - quality has to be managed, not left to chance.

TQM is a management approach of an organisation, centred on quality, based on the participation of all its members and aiming at long-term success through customer satisfaction and benefits to all members of the organisation and to society. (ISO 8402:1994)
TQM is both a **philosophy** and a set of guiding principles that represent the foundation of a continuously improving organisation. It integrates fundamental management techniques, existing improvement efforts and technical tools under a disciplined approach focused on continuous improvement. (Besterfield)

- Another simple definition: TQM = **TQC (Total Quality Control) + Quality Circles + Continuous improvement + Employee involvement.**

Evolution of quality (quick timeline)



- **Inspection** (1920s): sorting good from bad after production. Detect defects.
- **Quality control** (1930-50s): statistical methods, sampling, control charts (Shewhart).
- **Quality assurance** (1950-70s): prevent defects by system - documents, audits, quality manuals.
- **TQM** (1980s onwards): whole organisation, customer focus, culture of continuous improvement.

Difference: Traditional management vs TQM

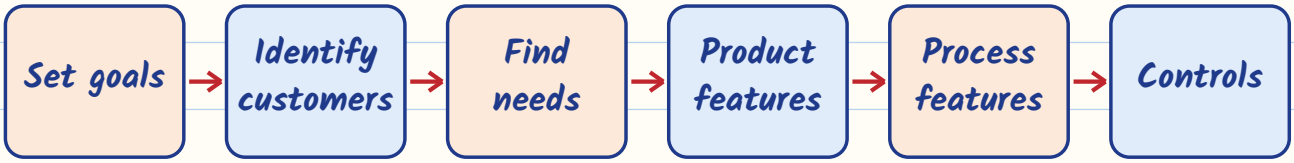
Aspect	Traditional	TQM
Focus	Product, profit	Customer satisfaction
Quality responsibility	QC department	Everyone
Approach	Detect and correct	Prevent defects
Improvement	Occasional	Continuous
Supplier	Adversary, many	Partner, few
Decision making	Intuition	Data and facts
Employees	Cost, followers	Asset, empowered teams

4. Quality Planning

Quality planning is the activity of establishing the quality objectives and requirements, and of defining the operational processes and related resources needed to fulfil them. (Juran: the first process of the Juran Trilogy)

Juran's steps of quality planning (6 steps)

1. **Establish quality goals** - what is to be achieved.
2. **Identify the customers** - internal and external.
3. **Discover customers' needs** - stated, real, perceived, cultural, unintended uses.
4. **Develop product features** that respond to the needs.
5. **Develop process features** capable of producing those features.
6. **Establish process controls** and transfer the plan to operations.



Quality planning for a product (practical stages)

- Market research - customer need, competitor study.
- Design and development - specifications, prototypes, design review.
- Process planning - machines, tools, methods, inspection plan.
- Supplier planning - material specs, vendor approval.
- Quality manual, work instructions, check sheets.
- Training and resource planning.
- Feedback and corrective action planning.

Benefits of quality planning

- Fewer design changes and customer complaints.
- Better use of resources, less waste and rework.
- Clear objectives, so everybody knows the target.
- Shorter time to market, lower life-cycle cost.
- Quality becomes proactive and not reactive.

5. Quality Costs

Cost of Quality (COQ) is the sum of costs incurred in preventing poor quality, appraising quality and the costs due to failure (internal and external). It is the cost of doing things wrong - and of ensuring they are done right.

Four categories of quality cost (PAF model)

Prevention cost → **Appraisal cost** → **Internal failure** → **External failure**

a) Prevention costs (cost of doing it right first time)

- Quality planning, training of employees, quality circles.
- Supplier evaluation, process control, preventive maintenance.
- Design reviews, pilot runs, quality improvement projects.

b) Appraisal costs (cost of checking)

- Incoming material inspection, in-process and final inspection.
- Testing, calibration of instruments, laboratory, quality audits.
- Cost of inspection staff and test equipment.

c) Internal failure costs (defects found BEFORE delivery)

- Scrap, rework/repair, re-inspection, retest.
- Downtime due to defects, yield losses, material review, failure analysis.

d) External failure costs (defects found AFTER delivery)

- Warranty claims, returns and replacements, complaint handling.
- Product recall, field service, product liability, penalties.
- **Loss of customer goodwill and future sales** (the hidden cost).

Prevention + Appraisal = **Cost of Good Quality (conformance).**
Internal + External failure = **Cost of Poor Quality (non-conformance).**

The iceberg

- Visible costs (scrap, rework, warranty) are only about **10 to 20%** of the real cost.
- Hidden costs: lost customers, lost sales, overtime, expediting, extra inventory, low morale, management time. These form the big part below the water.

Optimum quality cost model

- Prevention and appraisal cost **increase** as the quality of conformance (defect-free level) rises.
- Failure costs **decrease** as quality improves.
- Total cost curve is U-shaped - minimum at the optimum point. Modern view: spend more on prevention and the total cost keeps falling - 100% good is economical (zero defect view).

Quality level	Prevention + appraisal	Failure cost	Total

Low	Low	Very high	High
Optimum	Moderate	Moderate	Minimum
Very high (modern view)	Prevention emphasised	Very low	Still falling

- Typical figures: COQ is 15-30% of sales in poor organisations; world-class firms keep it below 5-10%.

6. Analysis of Quality Costs

Quality cost analysis tells where the money is being lost, so improvement effort can be targeted.

Steps of analysis

1. Decide the cost categories and identify the elements in each.
2. Collect data from accounts, inspection and rework records, warranty reports.
3. Prepare the **quality cost report** - category-wise total.
4. Compare with a **base** - sales, manufacturing cost, direct labour, units produced.
5. Study trends month to month and interpret.
6. Select projects with the highest cost (Pareto) and act.

Typical bases for comparison

- **Sales:** COQ / net sales x 100
- **Manufacturing cost:** COQ / total manufacturing cost x 100
- **Direct labour hours or cost,** and **units of product.**

Example - analysis table (Rs. lakh)

Category	Cost	% of total
Prevention	4	8%
Appraisal	10	20%
Internal failure	16	32%
External failure	20	40%

Total COQ	50	100%

- Reading: 72% is failure cost - firm is a "firefighter". Needs to move money to prevention.
- Rule of thumb: ideal ratio about Prevention 50-60%, Appraisal 20-25%, Failure 15-20% of COQ.
- Good indicators: failure cost falling, prevention rising, COQ % of sales falling.

7. Techniques for Quality Costs

- **Pareto analysis:** arrange cost elements in decreasing order; 80% of cost comes from 20% of causes - tackle the vital few.
- **Trend analysis:** plot quality cost against time, sales or production to check improvement.
- **Ratio analysis:** COQ as % of sales, of production cost, of labour, per unit.
- **Cause and effect / Fishbone diagram** to trace the reason for the highest failure cost.
- **Process cost approach:** (ASQ) cost of conformance + cost of non-conformance for each process.
- **ABC (activity based costing) of quality:** charge costs to the activities that cause them.
- **Taguchi loss function:** loss to society even within the specification (Unit IV).
- **Quality cost budgeting and control:** set target, compare actual, variance report.
- **Flow charts:** identify where the costs arise in the process.
- **Quality cost reports:** monthly management summary, departmental reports, project reports.

Uses / Benefits of quality cost analysis

- Gives a common language - money - to talk with top management.
- Identifies improvement opportunities and justifies investments in prevention.
- Measures the effect of the quality programme.
- Helps in budgeting and in resource allocation.

Limitations

- Hidden costs are difficult to measure; data collection is expensive.
- Needs a good accounting system; different definitions across firms.

8. Basic Concepts of TQM

> TQM rests on the idea that quality is built in during design and production, not inspected in at the end. It is a long term, company-wide effort.

Main elements

- **Customer focus:** the customer decides whether quality is good. Know the needs, measure satisfaction.
- **Leadership:** top management creates vision, values, policy and provides resources.
- **Employee involvement and empowerment:** teams, quality circles, suggestion schemes, training.
- **Continuous process improvement:** PDCA, Kaizen - small and never ending steps.
- **Process approach:** results come from controlled processes.
- **Supplier partnership:** suppliers are part of the quality chain.
- **Fact based decisions:** data, SPC and the seven tools.
- **Prevention not detection:** "do it right the first time" (Crosby).
- **Benchmarking and performance measures.**
- **Systems approach:** integrate all the processes towards one aim.

The TQM wheel/pillars



Principles in short

- Internal customer concept - each person receives from one and supplies another. "Next process is the customer".
- Zero defects, Right first time.
- Quality is a never-ending journey, not a destination.
- Cost of quality - spend to prevent, not to fix.
- Open communication, trust and teamwork across departments.

Obstacles (preview of Unit II)

- Lack of top management commitment, resistance to change, poor planning, wrong training, short-term thinking.

Key terms to remember

Term	Meaning
Quality assurance	Planned actions to give confidence that requirements are met
Quality control	Operational techniques to fulfil requirements
Quality circle	Small voluntary group solving work problems
Zero defects	Crosby's performance standard
Internal customer	Next person/department in process

Important Exam Questions (Unit I)

Short answer (2 marks)

- Define quality and TQM.
- List the 8 dimensions of quality.

- What are the SERVQUAL dimensions?
- Define cost of quality. What are the four components?
- What is the Pareto principle?

Long answer (10 marks)

1. Explain the dimensions of quality with suitable examples.
2. Define TQM. Explain the basic concepts and principles of TQM.
3. Explain quality planning. Discuss Juran's quality planning steps.
4. Describe the elements of quality cost. How are quality costs analysed? Explain the optimum cost model.
5. Compare the traditional management approach with TQM.

Revision line: Quality = customer satisfaction. TQM = everyone + everything + continuously. COQ = Prevention + Appraisal + Failure.